

Research paper

Augmenting Granger Causality through continuous ordinal patterns

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ABSTRACT

We here propose a novel methodology, based on the concept of continuous ordinal patterns, to preprocess time series and make explicit the non-linear temporal structures in them present. Through a series of synthetic and real-world examples, we show how such transformation overcomes one major limitation of the celebrated Granger Causality test, and allows to efficiently detect non-linear causality relations without the need of a priori assumptions. We further show how such transformation can be optimised based on the time series under study; but that good results can also be achieved using random ordinal patterns, in a way similar to how randomness is exploited in Reservoir Computing. We finally discuss the complementarity between this approach and the standard Granger one, especially in the analysis of real-world, and hence unknown, causal relations.

1. Introduction

One of the fundamental steps in the scientific analysis of real-world complex systems is the assessment of causal relations between their constituting elements [1], especially through the analysis of data representing their dynamics. This allows both to unveil the internal structure of the system under study; but also, and possibly more importantly, to design ways for steering its dynamics to a desired target [2]. Even though causality may not truly be detected in real data (as this usually requires an intervention on the system [3]), many proxy tests have been proposed in the last decades, of which the most well-known is probably the Granger Causality (GC) test [4]. A prototypical exponent of *predictive causality* [5], this test assesses whether the inclusion of information about the driving (i.e. causing) element helps predicting the future dynamics of the driven (caused) one. In spite of many advantages, including its conceptual and computational simplicity, this test also presents some drawbacks, the most important of which being its linear nature — i.e. only linear influences of an element on a second one can be detected.

In this contribution we explore whether the GC test can be augmented through the use of continuous ordinal patterns [6]. These are a recently proposed modification of the well-known ordinal pattern concept introduced by Bandt and Pompe [7], aimed at assessing how well a time series is described by a short pattern. In essence, such patterns can be seen as a way of quantifying, or making explicit, the presence of specific non-linear structures in short segments of the original time series; and, when applied to pairs of time series, they can further be used to detect similarity or difference patterns between them. As such, we hypothesise that they can be used as a way of preprocessing the information, and of making explicit non-linear relations that may not be detectable by the standard GC test.

In what follow, we start by reviewing the main methods used throughout the contribution, including the concept of continuous ordinal patterns (Section 2.1); how these can be optimised given one or more dynamical systems (Section 2.2); and finally, how they can be combined with the Granger Causality test (Section 2.3). Next, the proposed approach is tested through synthetic time series, starting from simple chaotic dynamics (Section 3.1 and following), to a more complex model of brain activity (Section 3.4). We further apply the methodology to a real-world problem, involving the detection of causal relationships in the propagation of air

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transport delays (Section 3.5). These examples illustrate how the proposed approach is able to substantially increase the sensitivity of the Granger test, provided the underlying relationships are non-linear; and how this can be achieved even using random patterns, thus not entailing a significant additional computational cost. Finally, some conclusions about the advantages and limitations of the use of continuous ordinal patterns are discussed in Section 4.

2. Methods

2.1. Continuous ordinal patterns

Continuous ordinal patterns, firstly introduced in Ref. [6], can be considered an extension of classical ordinal patterns, which are based in representing sub-windows of the time series under study through symbols encoding the amplitude ranking of its constituting elements [7]. By definition, these symbols are fixed — given a sub-window of length D , usually called the embedding dimension, the ranking of its values can only be given by one of $D!$ possible permutations. Continuous ordinal patterns turn this idea around: patterns are defined prior to the analysis, for then evaluating how well they represent a given sub-window of the data.

More specifically, let us consider a time series X composed of n values $X = (x_1, x_2, \dots, x_n)$; and a sub-window s_t of it, of size D and starting at time t , given by $s_t = (x_t, x_{t+1}, \dots, x_{t+D-1})$. In order to obtain comparable results for different time series, this sub-window is normalised in the range $[-1, 1]$, yielding a new sub-window $s_t^* = (x_t^*, x_{t+1}^*, \dots, x_{t+D-1}^*)$. Next, we suppose a continuous pattern π of D values normalised in the range $[-1, 1]$; this can be predefined based on some knowledge about the dynamical system under analysis, or can be extracted from the data themselves, as will be shown below. In order to understand how well the dynamics of s^* is represented by π , the distance between both is calculated as:

$$\phi_\pi(t) = \frac{1}{2D} \sum_{i=0}^{D-1} |\pi_i - s_{t+i}^*|. \quad (1)$$

Note that $\phi_\pi(t)$ is defined between zero and one, these two extreme representing cases in which π and s^* are identical or have a different structure, respectively. When averaged over all sub-windows of a time series, $1 - \langle \phi_\pi(t) \rangle$ (or $1 - \phi_\pi$ in short) quantifies how well the dynamics of the time series is represented by the considered continuous pattern.

The classical and continuous versions of ordinal patterns share some major features: they both analyse the relative amplitude of neighbouring values in a time series, thus capturing the internal causal structure; are resilient to observational noise; and their calculation entails a low computational cost. At the same time, they also present three major differences. Firstly, each sub-window is mapped into one of $D!$ permutation patterns in the classical case; into a real value $\phi(t)$ in the continuous case. Secondly, the classical case entails finding the permutation pattern that best describes a given sub-window; on the contrary, the continuous case starts with a predefined pattern π , and quantifies how well it represents the sub-window under analysis. Thirdly, classical permutation patterns are sensitive to the presence of equal values in the time series [8,9], which should be eliminated by, for instance, introducing a noise of small amplitude; this problem is nevertheless not present in the continuous case, as the calculation of Eq. (1) does not rely on rankings — see also Ref. [6] for a detailed discussion.

2.2. Optimising continuous ordinal patterns: the simulated annealing algorithm

As previously highlighted, the pattern π used to characterise the time series is here a parameter, as opposed to the result of the analysis itself. It is thus necessary to introduce a criterium to guide its definition. Beyond scenarios in which the target π is known a priori, two additional cases can be envisioned.

Firstly, one may want to find the best π to describe an individual time series; or, in other words, the aim is to find the π minimising the average (or median) ϕ_π . Secondly, it may be necessary to compare two time series, or to find the differences between them; in this case, the two probability distributions of ϕ_π , corresponding to the two time series, can be compared through a, e.g., Kolmogorov–Smirnov (KS) test, with the objective of finding the pattern π maximising the p -value. In both cases, this boils down to a minimisation problem.

In what follows, such best pattern is found using a Simulated Annealing (SA) optimisation algorithm [10], aimed at extracting the uniqueness of each time series. Specifically, and as will be detailed below, we need to analyse two time series representing systems A and B , for which we hypothesise that B is causing A . We thus use the SA algorithm to find the set of D values constituting the pattern π that maximise the p -value of a two-samples KS test on the two series $\phi_\pi^A(t)$ and $\phi_\pi^B(t)$, respectively obtained by applying the pattern π to the time series of A and B . In order to respect the restrictions previously described for π , its values are limited to the range $[-1, 1]$, with at least two values required to be -1 and 1 .

Note that other alternatives can also be considered. For instance, an exhaustive search could be performed, provided D is small enough. Alternatively, one could use random patterns; even if counterintuitive, using random transformations yields good results, as will be explored below.

2.3. Assessing the presence of causal relations

The Granger Causality (GC in short) test [4], developed by the economy Nobel Prize laureate Clive Granger on top of time series prediction models proposed by Norbert Wiener [11], is one of the best well-known metrics for assessing predictive causality [5] between elements composing a system. It is based on testing an intuitive assumption over pairs of time series A and B , for which we hypothesise that B is causing A : whether including information about the past of B helps to predict the future of A — or, in other words, whether B contributes in defining the future of A . This is usually done by comparing two autoregressive-moving-average (ARMA) models, forecasting the future dynamics of A by respectively including or not information about the past of B ; for finally performing an F-test on the residuals, and thus obtaining a p -value assessing whether the presence of information about B is relevant — and hence, whether a causality is present. Since its introduction, the GC test has become a standard tool in many fields of science, including economics [12,13], neuroscience [14,15], politics [16] and climate [17], to name but a few examples.

Given two time series, we here compare two scenarios: applying the GC test on the original time series; and applying it on the time series transformed through a continuous ordinal pattern π , i.e. on the time series given by ϕ_π^A and ϕ_π^B . The pattern π is chosen prior to the application of the GC step, thus avoiding overfitting; and, unless otherwise specified, is obtained by maximising the difference between the two time series through a SA algorithm — as discussed above. In other words, we use a pattern π to extract what makes the two time series unique, and the GC test is used to detect whether the unique dynamics of the two elements are causally connected. Note that the result yielded by the two scenarios is the same, i.e. the p -value of the GC test, the only difference being the data on which this is calculated. This ensures that no special statistical analysis is needed, and that the proposed method can seamlessly be integrated into existing data analysis pipelines.

For the sake of clarity, what here reported is the probability of correctly detecting a causal relationship, after applying a significance threshold $\alpha = 0.01$ and over a large number of independent realisations, with the hypothesis that the application of the pattern π improves the sensitivity of the test.

3. Results

3.1. Chaotic time series

We start by testing the proposed approach through a simple dynamical system, composed of a linear combination of two Logistic maps:

$$\begin{aligned} x(t) &= ax(t-1)[1-x(t-1)], \\ y(t) &= ay(t-1)[1-y(t-1)], \\ z(t) &= (1-\gamma)y(t) + \gamma x(t-1). \end{aligned} \quad (2)$$

x and y thus represent two independent chaotic dynamics (with $a = 4$); and z a linear mixture of both, in which the weight of the contribution of x is given by γ . Note that this last parameter defines the intensity of the causality (or coupling) between x and z , such that both time series are independent for $\gamma = 0$, while x completely defines the future of z for $\gamma = 1$. Being the relationship between x and z a linear one, the GC test should be able to correctly detect it, provided the coupling is large enough and the time series are long enough. Fig. 1(a), black line, reports the fraction of times the Granger Causality yields a statistically significant result (p -value below 0.01) for the causality $x \rightarrow z$, as a function of the coupling strength. As expected, a perfect result can be obtained for γ as small as 0.15. The two other solid lines of the same panel report the evolution of the same metric when the x and z time series are transformed through the proposed methodology. Results are in this case marginally better, with shorter patterns (i.e. $D = 3$, light blue line) outperforming longer ones ($D = 4$, purple line). Finally, the grey dashed line corresponds to results for $D = 3$ when the time series under analysis are randomly shuffled, thus destroying any temporal structure.

As an example of a more complex scenario, we next consider the following coupled systems:

$$\begin{aligned} x(t) &= \sin(2\pi t/100), \\ y(t) &= [0.5x(t-1) + 3.5] \times \\ &\quad y(t-1)[1-y(t-1)]. \end{aligned} \quad (3)$$

While also in this case y corresponds to a Logistic map, here x is a sinusoidal signal that modulates the dynamics of the map, by directly changing its parameter a — here equal to $0.5x(t-1) + 3.5$. The relationship between both systems is thus no longer linear: the driving time series defines whether the second one is in a chaotic regime or not. Due to this non-linear nature, the GC test is no longer able to reliably detect the causality $x \rightarrow y$ (see Fig. 1(b), black line); on the contrary, the proposed method almost achieves a 100% accuracy, even for low values of γ (light blue and purple lines of the same panel), without any impact on the number of false positive results (grey dashed line).

An interesting conclusion can be drawn at this stage. The non-linear transformation introduced through the continuous ordinal patterns is not significantly affecting the results when the causality relations are linear, as the GC is specifically designed to detect these. On the other hand, the proposed approach is able to make explicit any change in the dynamics, and thus allows the GC test to detect relationships that are outside its initial capabilities.

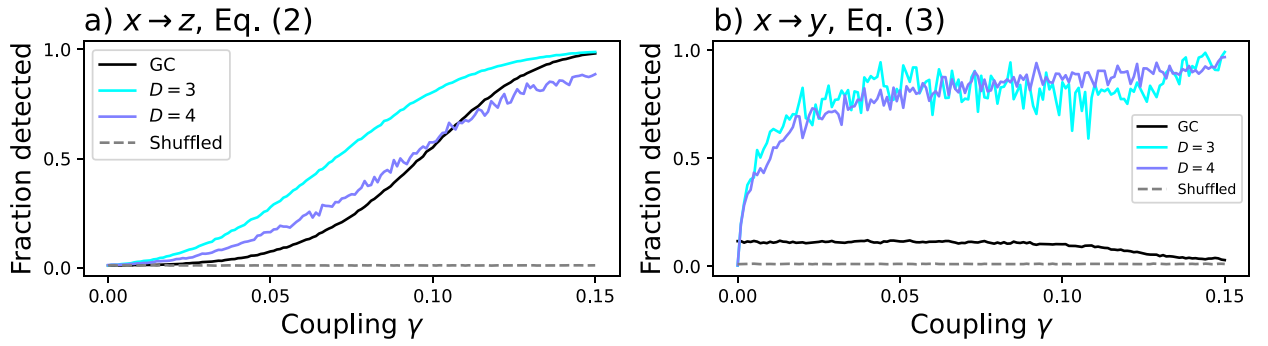


Fig. 1. Analysis of coupled chaotic systems. The top and bottom panels report the fraction of detected causal relationships, as a function of the coupling strength γ , respectively for the systems of Eqs. (2) and (3). Black lines correspond to the GC test, light blue and purple lines to the transformation using continuous patterns of size $D = 3$ and $D = 4$. Grey dashed lines correspond to results for $D = 3$ when the time series under analysis are randomly shuffled. Results obtained over 10^3 independent realisations, using time series of length 10^3 .

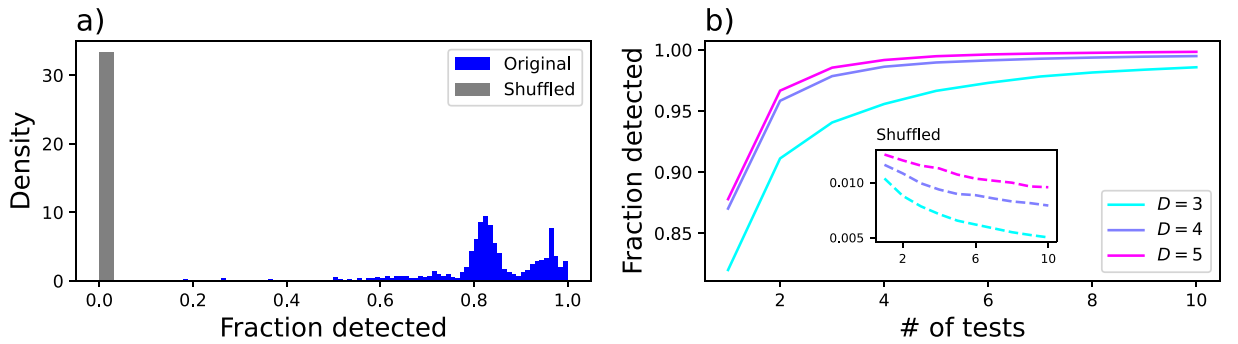


Fig. 2. Using random patterns π_{rnd} . (a) Histogram of the probability of finding a causal relationship in the systems of Eq. (3) (with $D = 3$ and $\gamma = 0.05$) when π is chosen at random. Blue and grey distributions respectively correspond to the original and randomly shuffled time series. (b) Evolution of the fraction of detected relationships when multiple tests with random patterns are performed and the best one is retained, as a function of the number of such tests. The inset reports the evolution of the same metric for randomly shuffled time series. All results correspond to 10^3 independent realisations, using time series of length 10^3 .

3.2. Role of the optimisation

We use the last introduced coupled systems to explore the importance of the SA optimisation procedure, and specifically whether it is required at all. Towards this aim, panel (a) of Fig. 2 reports the histogram (in blue) of the fraction of detected time series when the continuous ordinal pattern π is created at random (for the system of Eq. (3), with $D = 3$ and $\gamma = 0.05$). When compared with the results of Fig. 1(b), it can be appreciated that results are slightly worse than those obtained with the optimised pattern (mean of 0.828, median of 0.833). Still, the large majority of the causal relationships are detected, and the sensitivity is clearly above that of the original GC test. This does not come at the cost of a significant number of false positives, as indicated by the grey distribution for randomly shuffled time series.

This may *prima facie* come as a surprise: using any random pattern, and hence any random transformation, is better than not using one; and using an optimal one (at least, optimal according to the SA algorithm) only yields a marginal improvement. It is nevertheless not the first time a similar behaviour is observed, and it is instead the foundation of Reservoir Computing [18,19]. This paradigm is based on feeding the input data into a “reservoir” composed of a large number of (usually non-linear) dynamical systems, which are coupled according to a random topology. A subset of units is then read, and a (usually linear) model is fitted to recover the target of the computation. In other words, the input information is modified firstly by a random non-linear transformation, and secondly by a linear (and optimised according to a criterium) one. The computation here proposed is conceptually similar: the time series are firstly modified through the random pattern π_{rnd} , which corresponds to a random non-linear transformation; and are then processed by the linear GC test. A similar general principle may be buttressing both Reservoir Computing and the approach here presented: the non-linear transformation expands the input time series into a higher-dimensional space, which is easier to be explored by the subsequent linear model. This is also coherent with the result of Fig. 1(a): such expansion is unnecessary and yields no benefit when the relationship to be detected is already linear.

The results previously discussed suggest an interesting strategy, to be used in lieu of the SA algorithm: (i) calculate the p -value of the analysis with a number n of random patterns π_{rnd} ; (ii) select the test yielding the lowest p -value; and (iii) accept the presence of a causality is the p -value is below the significance threshold, after adjusting for the use of n tests. To illustrate, Fig. 2(b) reports the evolution of the fraction of detected causalities, for the coupled systems of Eq. (3) and $\gamma = 0.05$, as a function of the number

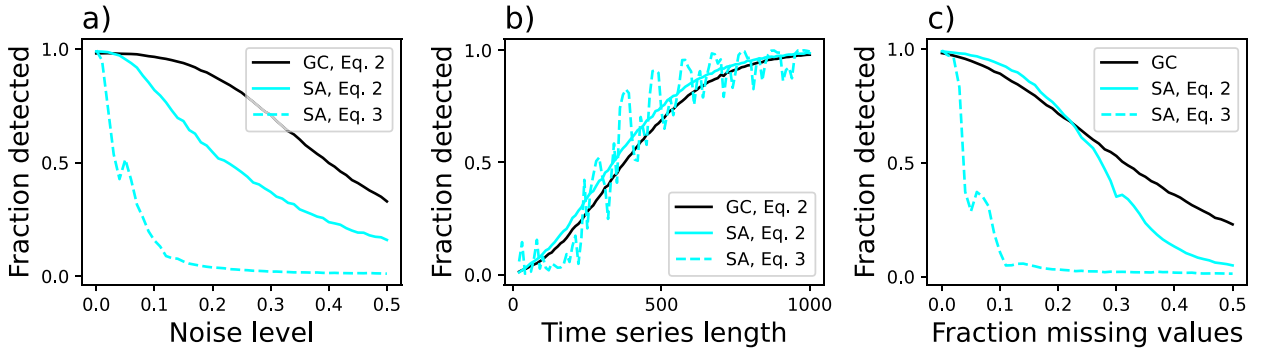


Fig. 3. Evolution of the fraction of detected causality links as a function of (a) the level of observational noise, (b) the length of the time series, and (c) the fraction of missing values. Black and light blue lines correspond to the standard Granger Causality and the continuous pattern version for the coupled systems of Eq. (2); the dashed light blue lines, to the continuous pattern version for Eq. (3). Time series length has been fixed to 10^3 in panels (a) and (c); and $d = 3$ and $\gamma = 0.15$ in all cases.

of tests performed in parallel, and with a Bonferroni correction of the p -value. An almost perfect result, and even superior to what reported in Fig. 1(b) using the SA algorithm, can be obtained with around ten tests. Note that this comes with the advantage of a substantially lower computational cost, as no computational intensive optimisation is required.

In order to ensure that this is not a spurious effect, i.e. that the detected causality relationships are not just the result of statistical fluctuations, we repeated the same analyses when the evaluated time series are randomly shuffled — i.e. deleting any causal structure. Results are reported through the grey distribution in panel (a) and the inset in panel (b) of Fig. 2. It can be appreciated that using one or multiple random patterns does not result in a significant number of causality relations, and hence of false positives, when the underlying data are random.

In synthesis, the proposed approach does not require a computationally costly optimisation procedure: in a way similar to how Reservoir Computing exploits randomness, good results can be obtained using one or more random continuous ordinal patterns π_{rnd} .

3.3. Noise, time series length, and missing values

As a last analysis performed with the synthetic time series generated by coupled Logistic maps, we here consider three aspects of relevance in the study of real-world time series. Firstly, the presence of observational noise, i.e. when independent random values from a distribution $\mathcal{N}(0, \sigma)$ are added to the time series $z(t)$ of Eq. (2) and to time series $y(t)$ in the case of Eq. (3). In both cases, the continuous pattern π is obtained by applying the SA algorithm over the noisy data. Results, in terms of the fraction of detected causal relationships, are reported in Fig. 3(a), for the original Granger Causality (black line); the continuous patterns one ($D = 3$, light blue solid line) for the systems of Eq. (2); and for the continuous patterns one in the case of the systems of Eq. (3) ($D = 3$, light blue dashed line). It can be appreciated that the introduction of π makes the analysis more sensitive to noise, with a significant decrease in the performance in the latter case. This is not surprising, as the non-linear nature of the continuous pattern is intrinsically more sensitive to noise — i.e., the effect of noise is amplified.

Secondly, we consider the role of the length of the analysed time series; Fig. 3(b) reports the fraction of detected relationships as a function of such length, using the same cases and colour code as panel (a). It can be appreciated that results are similar in the three cases, indicating that using the continuous pattern approach does not require longer time series. An oscillation of period 100 is also present in the case of Eq. (3) (light blue dashed line), corresponding to the period of the driving signal x .

Finally, panel (c) of Fig. 3 reports the same results, with the same colour code, as a function of the fraction of missing values in the time series. This analysis corresponds to what presented in Refs. [20,21], and to situations in which data cannot at times be recorded for either contingent or structural reasons. This is here simulated by setting a given fraction of values to zero. Results are qualitatively similar to the application of the original GC in the case of Eq. (2); but also show a marked decrease in the sensitivity of the test in the case of Eq. (3). This is once again to be expected, as the non-linear nature of the relationship of Eq. (3) increases the test sensitivity to errors in the data.

As a general guideline, when the relationship to be detected is linear, the application of the continuous pattern is not significantly affected by the properties of the time series under analysis. This is nevertheless not true in non-linear cases, for which the presence of observational noise or missing values has a negative impact. While reasonable, such limitation ought to be taken into account in the analysis of real-world data.

3.4. Synthetic brain data

As an example of the analysis of a synthetic model with a richer dynamics, we consider a biologically inspired dynamical system that emulates the signals recorded by functional Magnetic Resonance Imaging (fMRI) in the human brain [22]. The model is firstly based on generating coupled time series through a VAR model and additive Gaussian noise; secondly, on convolving them with the

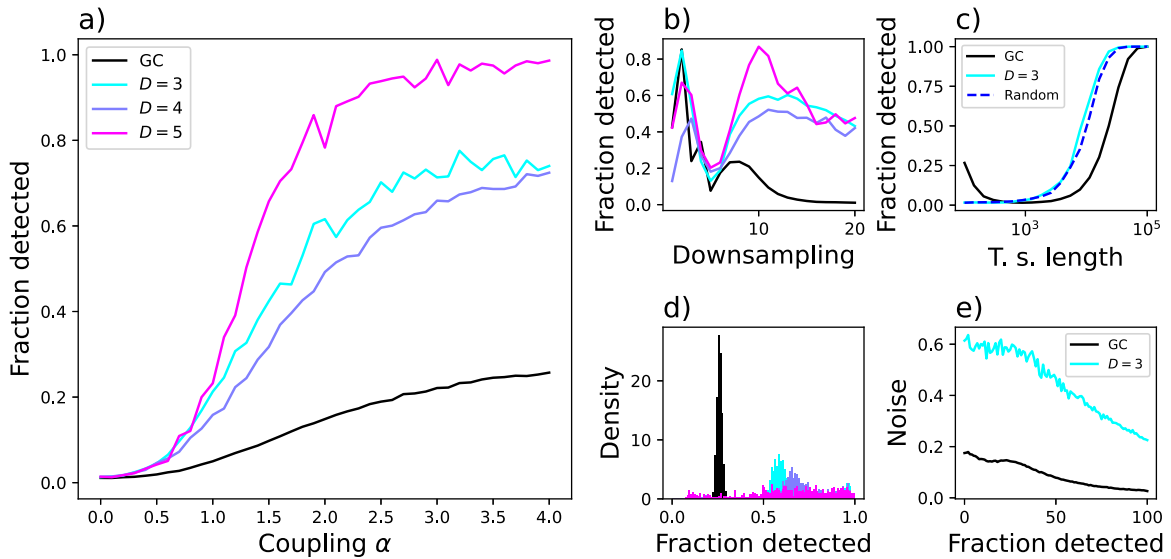


Fig. 4. Performance of the proposed approach on synthetic fMRI data. (a–c) Evolution of the fraction of detected causality relations as a function of the coupling α , of the downsampling, and of the time series length. (d) Probability distributions of the fraction of detected causalities for the GC, and for the proposed approach using random patterns π_{rnd} . (e) Fraction of detected causality relations as a function of the amplitude of a normally-distributed observational noise. Unless otherwise stated, results correspond to time series of length 10^4 , $\alpha = 2.0$, and a downsampling of 10. Results for the GC test, and for different values of the pattern length D , are colour coded as reported in the legends.

canonical haemodynamic response function (HRF), defined as a mixture of two gamma functions; and finally, on down-sampling the resulting output — for a full definition, please refer to Refs. [23,24]. While the time series yielded by this model are qualitatively similar to those observed in fMRI studies, the advantage is that here the coupling between them, i.e. what would correspond to a functional connectivity in the brain, can arbitrarily be defined and is therefore known.

Panel (a) of Fig. 4 reports the evolution of the fraction of detected causal relationships as a function of the coupling strength (see parameter α in Ref. [23]), for the original GC test (black line) and the proposed approach with patterns of sizes $D = 3$ (light blue line), 4 (purple line), and 5 (magenta line), optimised using the SA algorithm. Results indicate that the proposed approach significantly outperforms the original GC test, for reasons that will become clearer in the next panel.

Panel (b) of the same figure reports the evolution of the same metric as a function of the time series downsampling, for a fixed coupling of $\alpha = 2.0$. When the downsampling performed is small, the observed dynamics is dominated by a fast component that makes both time series highly correlated; consequently, the causal relation is mostly linear, and both approaches yield similar results. On the other hand, when such fast dynamics is deleted by using larger downsamplings, the causality becomes non-linear and is better detected by the proposed approach. Finally, for even larger downsampling, not enough information is retained and both methods fail in detecting any relationship. Notably, while using long patterns with $D = 5$ results in an improvement over $D = 3$, the same is not true for $D = 4$; the reasons behind such behaviour are not clear at this stage, and may be due to some characteristics of the synthetic time series.

Panel (c) of the same figure further reports the fraction of detected causality relations, for a downsampling of 10 and $\alpha = 2.0$, as a function of the final length of the time series — i.e. the length after the downsampling process. The proposed approach has lower requirements in terms of data quantity, and achieves results comparable to the original GC with up to one order of magnitude shorter time series. The blue dashed line further reports the result obtained using random patterns π_{rnd} , as previously discussed; this is only marginally worse, in exchange of a significant reduction in the computational cost. On a negative side, one has to keep in mind that using random patterns yields stochastic results, and hence that one test may not be enough to obtain a reliable assessment. This is especially evident in panel (d), reporting the probability distribution of the fraction of detected causalities when using random patterns; longer patterns yield more variable results, as they are sampled from a larger parameter space.

As a final point, panel (e) of Fig. 4 analyses the behaviour of the proposed approach as a function of the amplitude of an observational noise artificially added to both time series — with values drawn from a zero-mean normal distribution, as a function of its standard deviation. Once discounted the higher fraction of causal relations detected by the proposed approach in the case without noise, this and the original GC approaches seem to have comparable sensitivities.

3.5. Delay propagation in air transport

As a final case study, we consider the problem of detecting instances of delay propagations in air transport, i.e. when the late arrivals of aircraft to an airport induce delays in subsequent flights, and therefore in other airports [25]. The use of functional networks to describe the structure of such propagation has gained momentum in the last years [26–32], especially thanks to the

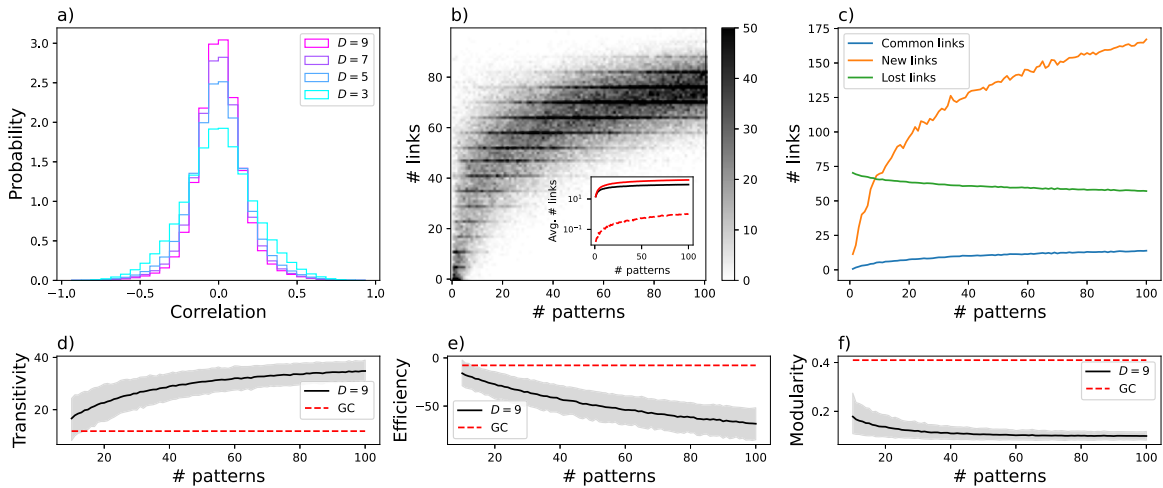


Fig. 5. Analysis of delay causality links in air transport. (a) Distribution of the rank correlations between the p -values obtained by the GC test using 500 different random patterns. The four curves correspond to four embedding dimensions D , see legend. (b) Probability distribution of the number of statistically significant link (with a Bonferroni correction) as a function of the number of random patterns. For each number of patterns, results correspond to 1000 independent analyses. The inset further shows the evolution of the average number of detected links with and without Bonferroni correction (respectively solid black and red lines); and the average number of links when the original time series are randomly shuffled, without Bonferroni correction (dashed red line). (c) Evolution of the number of links common to both using the original GC and the proposed approach (blue line); only detected by the latter (orange line); and only detected by the former (green line), as a function of the number of random patterns π_{rnd} . (d)–(f) Evolution of the Z-Score of the transitivity, efficiency, and modularity of the networks reconstructed through the GC approach (red dashed lines), and through the proposed approach as a function of the number of random patterns π_{rnd} (black solid lines, average, and grey bands, 10–90 percentile).

need of only macro-scale data and the easy interpretability of results. In this case, instances of delay propagation between pairs of airports are detected by applying correlation or causality tests on the time series representing the evolution of delays in those airports — hence describing whether the increase of delays at a given airport is causing a similar reaction in a second one. When such analysis is repeated for all pairs of airports in a system, the detected propagations are represented as links in a complex network [33,34], whose structure can be analysed through suitable topological metrics [35].

The analysis here proposed is based on data about air transport operations extracted from the EUROCONTROL's R&D Data Archive, a public repository of historical flights made available for research purposes, and freely accessible at <https://www.eurocontrol.int/dashboard/rnd-data-archive>. We have specifically considered the landing delay of all flights landing in the 50 largest airports in Europe, ranked according to their number of landing operations, for the month of September 2019; such delay has been estimated as the difference between the actual (from the ATFM-updated flight plan) and the planned (according to the last filed flight plan) landing times. For each airport, a time series representing the average landing delay within non-overlapping and consecutive windows of 20 minutes has been calculated [36]. In order to delete weekly and daily periodicities [37], the difference between the value obtained in a window, and the average of the delay in all other windows of the same week day and hour has been calculated. In other words, the final time series represent how the observed delay deviated from what expected in similar days and time periods. Finally, both the original GC test and the continuous pattern version have been applied to all pairs of airports, with a maximum lag of six hours.

Note that this analysis is substantially different from the previous cases. First of all, by being based on the evaluation of real time series, the underlying true causality is not known and its strength cannot artificially be tuned. Secondly, the focus shifts from the analysis of a pair of time series, to the evaluation of a full functional network — in this case composed of 50 nodes, and hence of 50×49 potential links. At the same time, it provides an excellent test case for understanding how a real data set can be studied, especially regarding the use of random patterns.

We start by analysing how the result yielded by each random pattern π_{rnd} depends on the pair of airports under analysis. For that, we firstly create vectors by applying 500 random patterns to the time series of a random pair of airports, for then extracting the corresponding p -value by applying the GC test on the transformed time series. Finally, the Spearman's rank correlation is calculated between all pairs of vectors, and the resulting distribution is reported in panel (a) of Fig. 5. Most correlations are close to zero, thus indicating that the pattern that may work for detecting the causality between a given pair of airports may not work for another pair — or, equivalently, that each link should be analysed independently.

We next use a set of random patterns with $D = 9$ to reconstruct the full functional network, for each link selecting the lowest p -value among those yielded by the GC using these random patterns, and by evaluating the statistical significance through a Bonferroni correction. Panel (b) of Fig. 5 reports the probability distribution of the number of detected links, as a function of the number of patterns considered. Results indicate a constant trend: the more random patterns are evaluated, the larger the number of significant causalities. At the same time, a large variability is present: for instance, using 50 random patterns yields between 40 and 90 links, depending on the patterns themselves. Such variability may be reduced by resorting to larger sets of random patterns; yet, this

decreases the significance threshold due to the Bonferroni correction, and thus leads to a saturation in the number of detected links. As a solution, we discarded such correction, noting that the tests for a given pair of airports are not independent as they are based on the same time series. This leads to a significant increase in the number of detected links (see inset in the same panel). At the same time, when randomly shuffling the original time series, the average number of spurious causality links remains below one, thus confirming that not using a correction for multiple testing is not increasing the number of false positive results.

We finally analyse how the functional networks obtained through the original GC test on one hand, and that obtained using the proposed approach on the other, compare. Panel (c) of Fig. 5 reports the evolution, as a function of the number of random patterns used in the analysis, of the number of shared links (blue line); of new ones, only detected by the proposed approach (orange line); and of links only detected by the standard GC (green line). It can be appreciated that links of the first group are a minority, independently on the number of random patterns used in the analysis. In the absence of further information it can be speculated that the underlying propagation process can both assume a linear and non-linear form [38]; and that therefore each approach is only able to detect a subset of all existing causality links. If this hypothesis is correct, a full analysis of delay propagation requires considering links detected with both methods, to include both linear and non-linear propagation instances. The fact that different types of links are unveiled is naturally impacting the structure of the reconstructed network. To illustrate this, panels (d) to (f) of Fig. 5 report the evolution, as a function of the number of random patterns used in the analysis, of the Z-Score of three classical complex network metrics: transitivity [39], efficiency [40], and modularity [41,42] — see references for detailed descriptions. The Z-Score is here used to get a normalised version of the metrics independently of the number of links in each network, as discussed in Ref. [43]. As may be expected, values of metrics obtained with the proposed method (black solid lines) are different from those yielded by the GC test (red dashed lines), suggesting that linear and non-linear propagations follow different structures.

4. Discussion and conclusions

In this contribution we have shown how the recently proposed approach of continuous ordinal patterns [6] can be used to preprocess pairs of time series, with the aim of improving the detection capability of the classical Granger Causality test. These patterns help to make explicit the non-linear structures present in the data, focusing on how adjacent values are ordered in amplitude. Such information is then available to be detected even by linear methods, thus providing a computational simple way of overcoming the linearity limitation of the Granger test. While it is not the first time ordinal patterns have been proposed as building blocks of causality tests [44–48], to the best of our knowledge this is the first instance of an approach using them as a preprocessing step; hence, all advantages of the Granger test are inherited, and the proposed approach can easily be integrated into existing data analysis pipelines. This is made possible by the fact that the continuous ordinal patterns analysis yields new time series consisting of continuous values, as opposed to the symbolic values of the classical approach; in the latter case, the only solution would be to rely on causality metrics based on discrete random variables, as e.g. Transfer Entropy [49].

As opposed to standard ordinal pattern analyses, here the pattern π used to preprocess the time series has to be defined beforehand. In the absence of a priori information about the system's dynamics, we here derive it through an optimisation process, based on the well-known SA algorithm. While the computational cost may in principle be a barrier, we have also shown that similar (if not better) results can be obtained by using one or more random patterns and by selecting the most statistically significant result, in a way similar to how randomness is exploited in Reservoir Computing [18,19].

Reported results confirm that the proposed approach is able to render explicit non-linear relationships in the data generated by different systems, from chaotic maps, to synthetic and real systems. Still, it has to be noted that this is not an “one-fits-all” approach, and that it entails some limitations. Firstly, even if the computational cost is minor when using random patterns π_{rnd} , a large number of independent patterns have to be evaluated; the analysis of very long time series can therefore be challenging. Secondly, while the best performing pattern π can yield some information about the underlying non-linear causal structure, its interpretation is not straightforward, and is indeed more complex than the interpretation of the linear coefficients returned by the GC analysis. Finally, when the causal relationship to be detected is already linear, applying the proposed transformation does not increase the accuracy of the GC test, and may even reduce it (see Fig. 1). This is to be expected, as such transformation implies making more complex a relationship that was initially simple and well-suited for the test. Along this line, it is trivial to envision situations in which the proposed approach yields no benefits. To illustrate, one can imagine two time series X and Y , where elements from the former are drawn from a given probability distribution (e.g. $x(t) = \mathcal{N}(0, 1)$), and the latter is constructed by shifting in time elements of the former (i.e., $y(t) = x(t-1) + \xi$, with ξ representing additive noise). Applying ordinal patterns would yield no additional information, as each value is independent from the preceding and following ones; in other words, causality is not encoded in the dynamics of the two time series. This suggests that an optimal approach entails the application of both the standard GC test and the modified one, in order to extract respectively linear and non-linear relationships. As shown in the case of air transport delays, this can be used to elucidate their nature. To illustrate, if the GC test yields better results, i.e. a smaller p -value, it can be concluded that the underlying relationship is mainly linear; on the other hand, those relationships only detected by the proposed approach must be non-linear, and/or implying a change in the underlying dynamics.

CRedit authorship contribution statement

Massimiliano Zanin: Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Massimiliano Zanin reports financial support was provided by European Research Council.

Data availability

Links to the data are provided in the manuscript.

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